EXPERIMENT NO: 6

Student Name: Divas Kumar
UID: 23BCS13735
Branch: BE-CSE
Section/Group: KRG-1A
Semester: 6th
Date of Performance: 26/02/2026
Subject Name: System Design
Subject Code: 23CSH-314

Aim

To design a highly scalable video streaming platform similar to YouTube that supports heavy video uploading, adaptive bitrate streaming, and real-time social interactions with high availability and low latency.

Objectives

- Understand the architecture of a massive-scale media streaming system.
- Identify functional requirements such as direct-to-cloud uploads and personalized feeds.
- Identify non-functional requirements including CDN caching, high throughput, and latency constraints.
- Analyze the decoupling of synchronous API requests using event-driven stream processing.
- Design REST APIs for video management, search, and user interactions.

Procedure

1. Studied real-world video streaming architectures and content delivery networks.
2. Identified core entities such as Users, Videos, Interactions, and View Histories.
3. Analyzed the Pre-Signed URL pattern to bypass API Gateway bottlenecks for large file uploads.
4. Collected functional and non-functional requirements for media delivery.
5. Designed APIs for video processing, metadata retrieval, and social engagement.
6. Evaluated streaming protocols (HLS/DASH) and separated thumbnail storage for optimization.
7. Analyzed stream processing (Apache Flink/Kafka) for real-time, high-volume view aggregation.

Functional Requirements

1. System should allow users to register, log in, and manage accounts.
2. System should support uploading large raw video files efficiently.
3. System should stream video smoothly using Adaptive Bitrate Streaming based on network conditions.
4. System should support real-time social interactions (likes, dislikes, comments).
5. System should allow users to search for videos by title or metadata.

6. System should generate a personalized home page feed based on watch history.

Core Entities of the System

- Users
- Videos
- Interactions (Likes/Dislikes)
- Comments
- ViewHistory

API Design

1. Authentication API

A. **Register**: POST /api/v1/auth/register
B. **Login**: POST /api/v1/auth/login

2. Video & Feed API

A. **Initialize Upload**: POST /api/v1/videos/upload/init (Returns S3 Pre-Signed URL)
B. **Get Video Metadata**: GET /api/v1/videos/{video_id}
C. **Get Personalized Feed**: GET /api/v1/feed (Pagination Supported)
D. **Search Videos**: GET /api/v1/search?q={query}

3. Interaction API

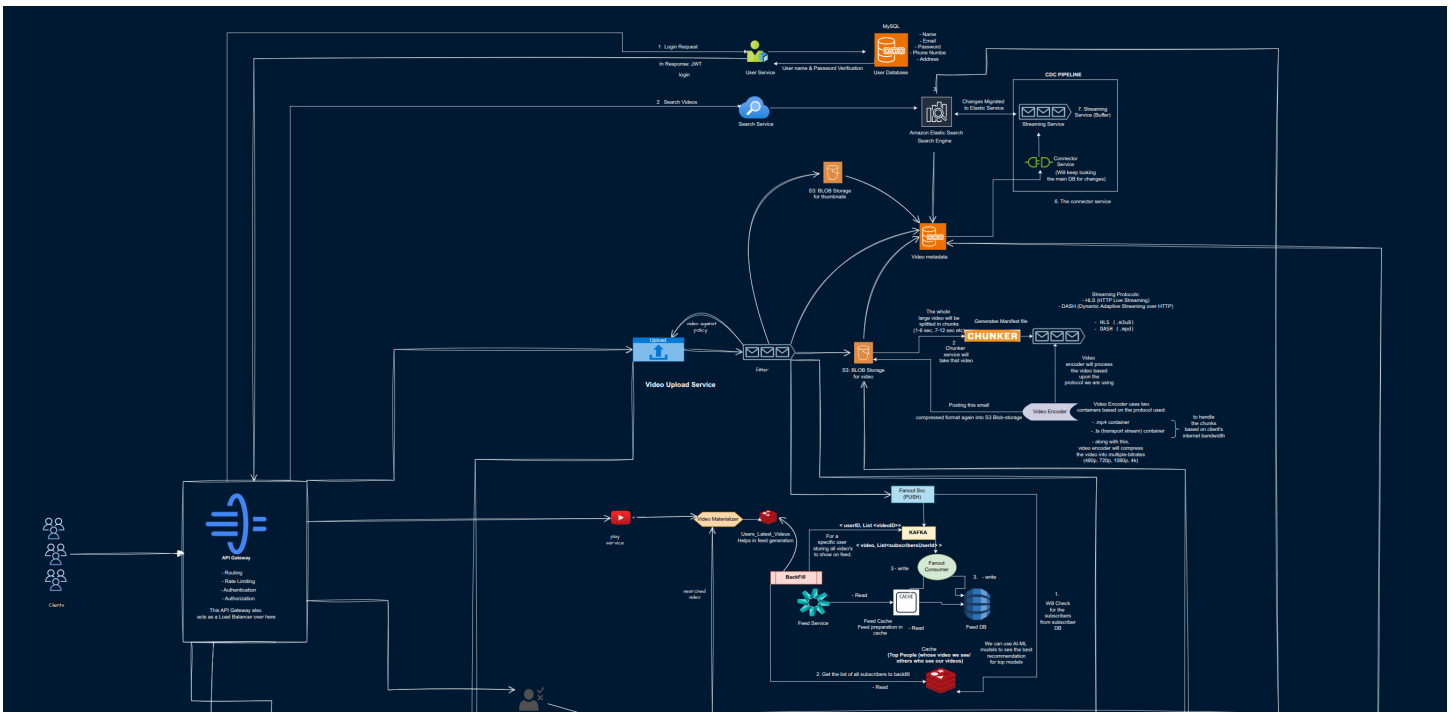
A. **Like Video**: POST /api/v1/videos/{video_id}/like
B. **Add Comment**: POST /api/v1/videos/{video_id}/comments
C. **Get Comments**: GET /api/v1/videos/{video_id}/comments

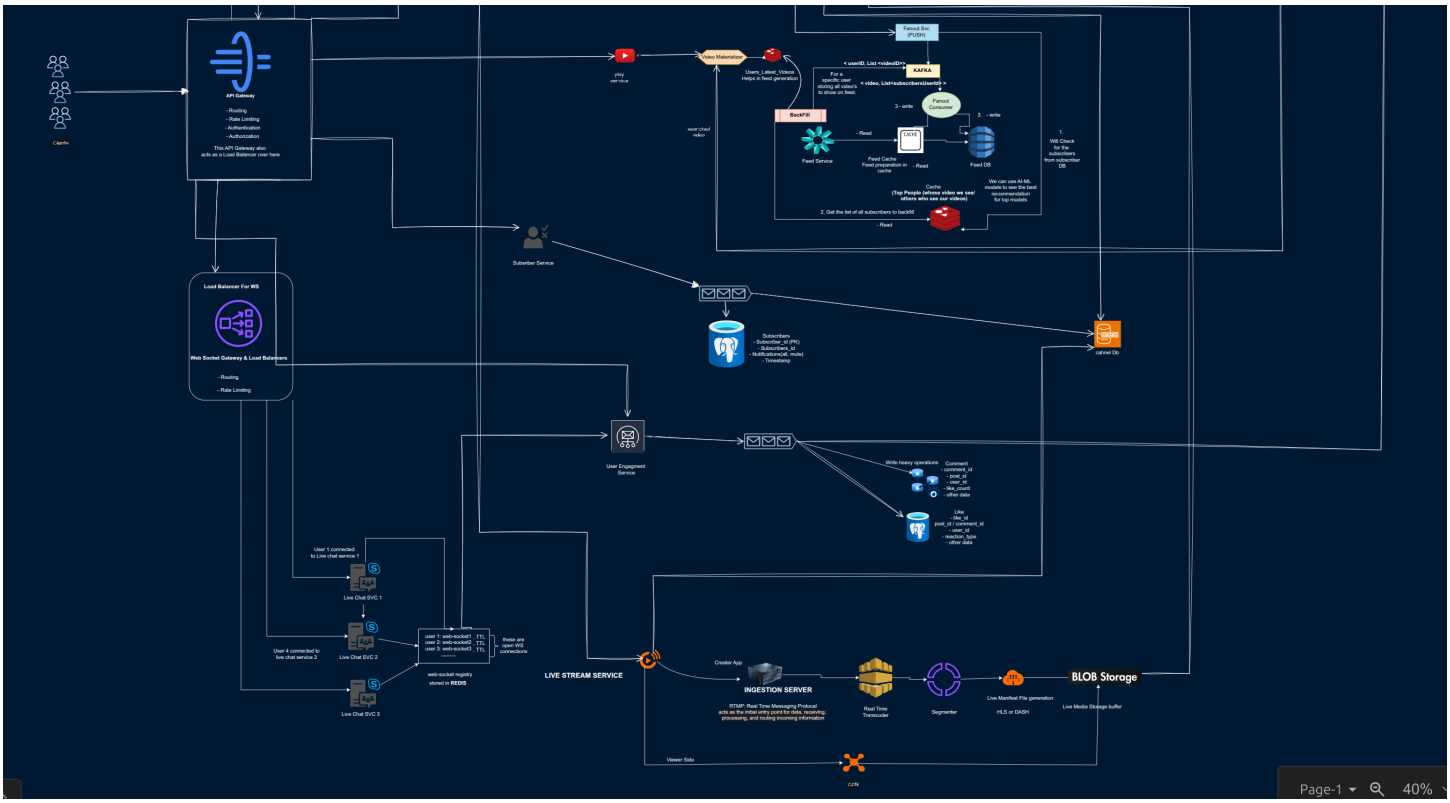
Non-Functional Requirements

1. System must handle extremely high throughput for concurrent uploads and millions of simultaneous views.
2. High Availability must be prioritized, ensuring no single point of failure.
3. System should follow an Eventual Consistency model for social metrics (view counts, likes) to optimize database write loads.
4. Video delivery latency should be minimized via Edge Caching and a Global CDN.
5. Transcoding pipeline must be horizontally scalable to handle viral upload spikes.

High Level Design (HLD)

The system consists of Client Applications, an API Gateway, and specialized microservices (Auth, Upload, Video Metadata, Feed, and Interactions). Heavy uploads bypass the backend via Pre-Signed URLs directly to Object Storage. Asynchronous transcoding pipelines process videos into HLS/DASH segments, which are delivered globally via a CDN. High-volume interactions are managed using Redis (Likes), NoSQL clusters (Comments), and a Kafka-to-Apache Flink stream processing pipeline for accurate view aggregation.


MySQL
- Name
- Email
- Password
- Phone Number
- Address
User Database
1. Login Request
In Response: JWT
login
User Service
User name & Password Verification
2. Search Videos
Search Service
Amazon Elastic Search
Search Engine
Changes Migrated to Elastic Service
CDC PIPELINE
7. Streaming Service (Buffer)
Streaming Service
Connector Service
(Will keep looking The main DB for changes)
6. The connector service
S3: BLOB Storage for thumbnails
Video metadata
Streaming Protocols:
HLS (HTTP Live Streaming)
- DASH (Dynamic Adaptive Streaming over HTTP)
- HLS (.m3u8)
- DASH (.mpd)
The whole large video will be splitted in chunks (1-4 sec, 7-12 sec etc)
Generates Manifest file
CHUNKER
2. Chunker service will take that video
Video encoder will process the video based upon the protocol we are using
Video Encoder
Video Encoder uses two containers based on the protocol used:
- .mp4 container
- .ts (transport stream) container
- along with this, video encoder will compress the video into multiple bitrates (480p, 720p, 1080p, 4k)
to handle the chunks based on client's internet bandwidth
Pushing this small compressed format again into S3 Blob-storage
video signed policy
Upload
Video Upload Service
Filter
S3: BLOB Storage for video
Clients
API Gateway
- Routing
- Rate Limiting
- Authentication
- Authorization
This API Gateway also acts as a Load Balancer over here
play service
Video Materalizer
Users_Latest_Videos
Helps in feed generation
watched video
Fanout Svc (PUSH)
< userID, List <videoID>>
KAFKA
For a specific user storing all video's to show on feed.
< video, List<subscribersUserId> >
Fanout Consumer
3. write
BackFill
Feed Service
- Read
CACHE
Feed Cache
Feed preparation in cache
Feed DB
1. Will Check for the subscribers from subscriber DB
We can use AI-ML models to see the best recommendation for top models
Cache
(Top People (whose video we see/ others who see our videos)
2. Get the list of all subscribers to backfill


Clients
API Gateway
- Routing
- Rate Limiting
- Authentication
- Authorization
This API Gateway also acts as a Load Balancer over here
play service
Video Materalizer
Users_Latest_Videos
Helps in feed generation
watched video
Fanout Svc (PUSH)
< userID, List <videoID>>
KAFKA
For a specific user storing all video's to show on feed.
< video, List<subscribersUserId> >
Fanout Consumer
3. write
BackFill
Feed Service
- Read
CACHE
Feed Cache
Feed preparation in cache
Feed DB
1. Will Check for the subscribers from subscriber DB
We can use AI-ML models to see the best recommendation for top models
Cache
(Top People (whose video we see/ others who see our videos)
2. Get the list of all subscribers to backfill
Subscriber Service
Subscribers
- Subscriber_id (PK)
- Subscribers_id
- Notifications(all, mute)
- Timestamp
cahnel Db
Load Balancer For WS
Web Socket Gateway & Load Balancers
- Routing
- Rate Limiting
User Engagement Service
Comment
- comment_id
- post_id
- user_id
- like_count
- other data
Like
- like_id
- post_id / comment_id
- user_id
- reaction_type
- other data
User 1 connected to Live chat service 1
Live Chat SVC 1
User 4 connected to live chat service 2
Live Chat SVC 2
Live Chat SVC 3
user 1: web-socket1 , TTL
user 2: web-socket2 , TTL
user 3: web-socket3 , TTL
These are open WS connections
web-socket registry stored in REDIS
LIVE STREAM SERVICE
Creator App
INGESTION SERVER
RTMP: Real Time Messaging Protocol acts as the initial entry point for data, receiving, processing, and routing incoming information
Real Time Transcoder
Segmenter
Live Manifest File generation
HLS or DASH
BLOB Storage
Live Media Storage buffer
Viewer Side
CDN
Page-1
40%

Outcome

- Designed a highly scalable, event-driven video streaming architecture.
- Identified critical functional and non-functional requirements for media delivery.
- Designed REST APIs and decoupled asynchronous processing pipelines.
- Understood the application of CDNs, adaptive bitrate streaming, and stream processing for massive-scale systems.